



Aalto University
School of Business

Section 7

Credit risk analysis

“A man goes bankrupt gradually, then suddenly.”

--Ernst Hemingway

Learning objectives

After studying this chapter, you will understand

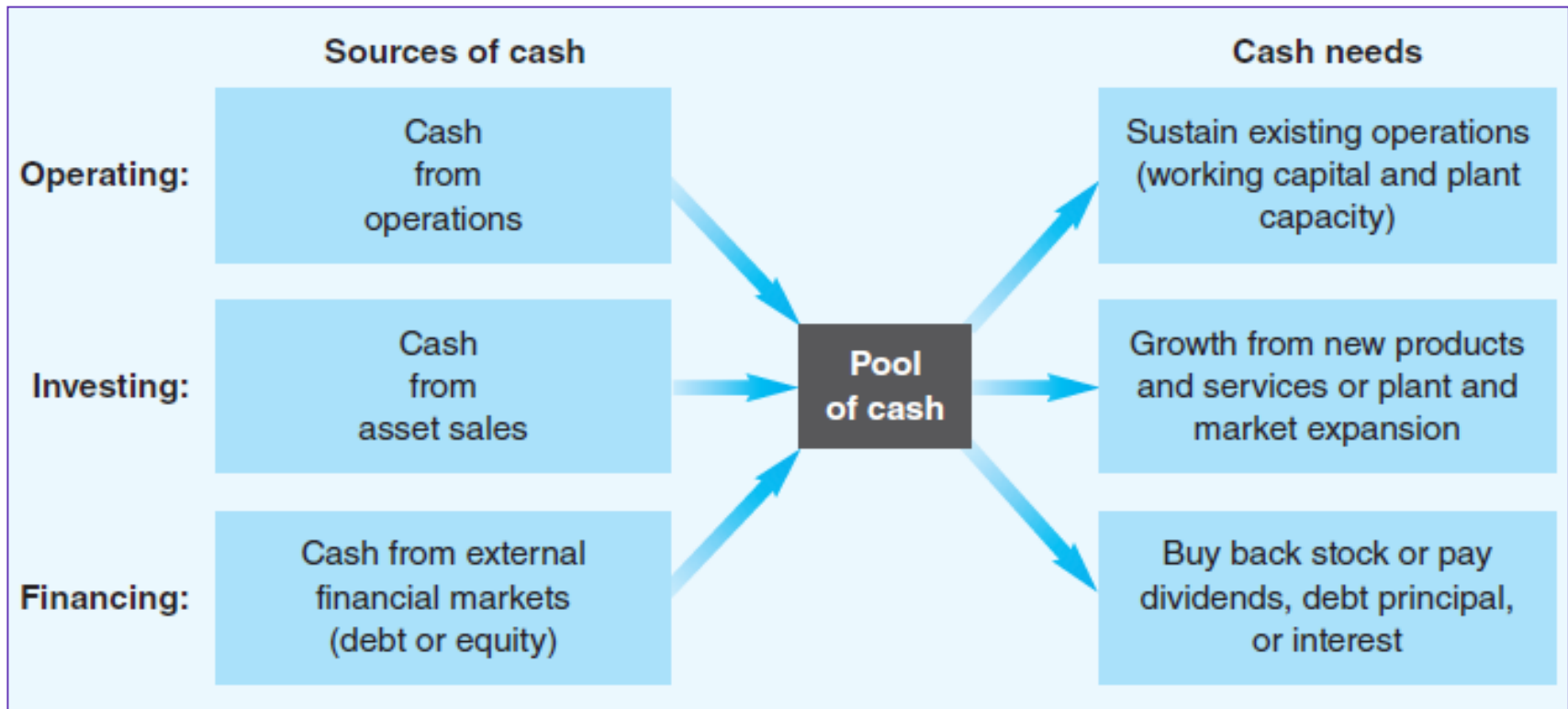
- A typical process of the financial distress of the firm
- How financial ratios can be used to predict financial distress and credit risk
- How other information of the firm helps predicting distress and credit risk

Key concepts

- Credit risk refers to the **risk of default** by the borrower
- The lender risks losing **interest payments** and **loan principal**
- A company's **ability to repay** debt is determined by its capacity to generate cash from operations, asset sales, or external financial markets in excess of its cash needs.
- A company's **willingness to repay** debt depends on which of the competing cash needs management believes is most pressing at the moment



Firm has to balance between cash sources and needs



Credit analysis

Step 1:

**Understand
the business**

- Business model and strategy
- Key risks and success factors
- Industry competition

Step 2:

**Evaluate
accounting quality**

- Spot potential distortions
- Adjust reported numbers as needed

Step 3:

**Evaluate current
profitability and health**

- Examine ratios and trends
- Look for changes in profitability, financial conditions, or industry position

Step 4:

**Prepare “pro forma”
cash flow forecasts**

- Develop financial statement forecasts
- Assess financial flexibility

Step 5:

Due diligence

- Kick the tires

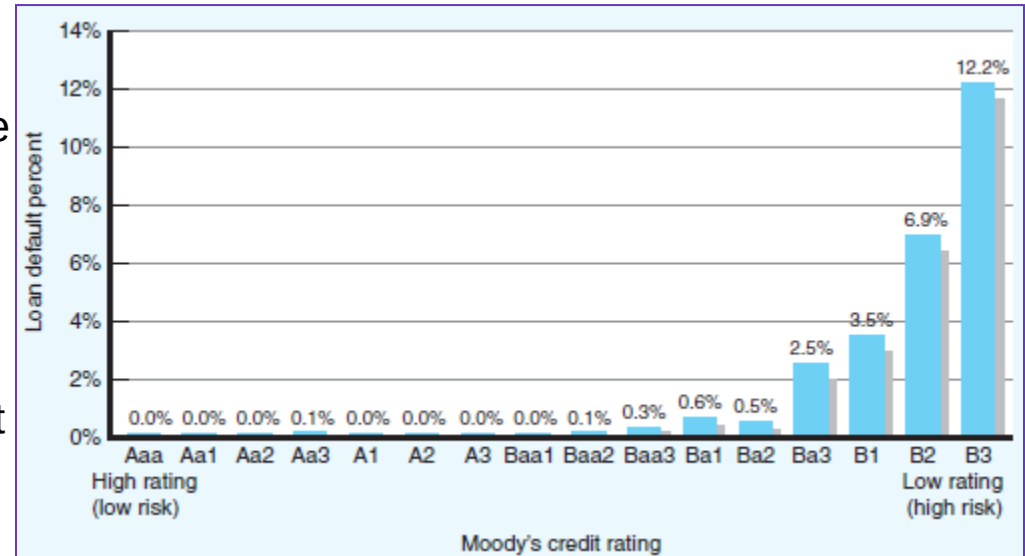
Step 6:

**Comprehensive risk
assessment**

- Likely impact on ability to pay
- Assess loss if borrower defaults
- Set loan terms

Financial ratios and credit risk

- A firm **defaults** when it fails to make principal or interest payments
- Lenders can then
 - Adjust the loan payment schedule
 - Increase the interest rate and require loan collateral
 - Seek to have the firm declared insolvent
- Financial ratios play two roles in credit analysis
 - They help quantify the borrower's credit risk before the loan is granted.
 - Once granted, they serve as an early warning device for increased credit risk



Default rates by Moody's credit rating

Source: Moody's Investors Service

Liquidity

- Liquidity refers to the ability of a firm to meet its **short-term financial obligations** when they fall due
- **Quick ratio** and **Current ratio** are commonly used liquidity measures
- Both ratios contain the assets in the numerator to include items that potentially can be converted into cash
- The higher both the ratios, the higher the liquidity position of the firm

Current ratio

- Frequently used measure of liquidity
- Assumes that current assets have a **liquidation value**, i.e. the firm could sell all its current assets to pay back its current liabilities, if it runs out its business
- Is calculated as follows (without time subscripts):

$$\text{CURRENT RATIO} = \frac{\text{CURRENT ASSETS}}{\text{CURRENT LIABILITIES}}$$

- Some guidelines to asses the level of Current ratio:

CR > 2	Good
1 <= CR <= 2	Satisfactory
CR < 1	Weak

Case: Current ratio of Kone, 2013

$$\begin{aligned}\text{CURRENT RATIO}_{2013} &= \frac{\text{CURRENT ASSETS}_{2013}}{\text{CURRENT LIABILITIES}_{2013}} \\ &= \frac{3405,0}{3217,4} \\ &= \mathbf{1,06}\end{aligned}$$

Quick ratio

- Another frequently used measure of liquidity
- Assumes that inventories do not necessarily have any liquidation value, if they are sold
- Is calculated as follows (without time subscripts):

$$\text{QUICK RATIO} = \frac{\text{CURRENT ASSETS} - \text{INVENTORIES}}{\text{CURRENT LIABILITIES}}$$

- Some guidelines to asses the level of Quick ratio:

QR > 1	Good
0,5 <= QR <= 1	Satisfactory
QR < 0,5	Weak

Case: Quick ratio of Kone, 2013

$$\begin{aligned} \text{QUICK RATIO}_{2013} &= \frac{\text{CURRENT ASSETS}_{2013} - \text{INVENTORIES}_{2013}}{\text{CURRENT LIABILITIES}_{2013}} \\ &= \frac{3405,0 - 1103,9}{3217,4} \\ &= 0,72 \end{aligned}$$

Net working capital ratio

- Measures the difference between current assets and current liabilities
- Net working capital is OFTEN defined as follows (without time subscripts):

NET WORKING CAP. = CURRENT ASSETS – CURRENT LIABILITIES

- Net working capital ratio is calculated as follows:

$$\text{NET WORKING CAPITAL RATIO} = \frac{\text{NET WORKING CAPITAL}}{\text{SALES}}$$

Case: Net working capital ratio of Kone, 2013

Current assets	3405,0
<u>Current liabilities</u>	<u>-3217,4</u>
= NET WORKING CAPITAL	187,6

$$\begin{aligned}\text{NET WORKING CAPITAL RATIO}_{2013} &= \frac{\text{NET WORKING CAPITAL}_{2013}}{\text{SALES}_{2013}} \\ &= \frac{187,6}{6932,6} \\ &= 2,7\%\end{aligned}$$

Turnover ratios

- These ratios measure how **effectively** a firm uses the components of net working capital
 - A firm should **minimize** the amount of working capital
 - Reduces invested capital, and hence, increases EVA
 - On the other hand, a firm must meet all its obligations when they are due
 - Frequently used ratios
 - Account receivables turnover rate
 - Account payables turnover rate
 - Inventory turnover rate
-

Account receivables turnover rate

- How quickly (or slowly...) the firm receives its receivables from sales

$$\text{ACCOUNT RECEIVABLES TURNOVER} = \frac{\text{SALES}}{\text{ACCOUNT RECEIVABLES}}$$

- Kone, 2013:

$$\begin{aligned}\text{ACCOUNT RECEIVABLES TURNOVER}_{2013} &= \frac{6932,6}{1164,6} \\ &= 6,0\end{aligned}$$

Account payable turnover rate

- How quickly (or slowly...) the firm pays its bills

$$\text{ACCOUNT PAYABLE TURNOVER} = \frac{\text{PURCHASES}}{\text{ACCOUNT PAYABLES}}$$

- Kone, 2013:

These numbers are from the footnote #5 in the I/S of Kone

$$\begin{aligned}\text{ACCOUNT PAYABLE TURNOVER}_{2013} &= \frac{2839,0 + 577,4 + 489,3}{511,2} \\ &= 7,6\end{aligned}$$

Inventory per sales-%

- How effectively inventory management works

$$\text{INVENTORY PER SALES-\%} = \frac{\text{INVENTORY}}{\text{SALES}}$$

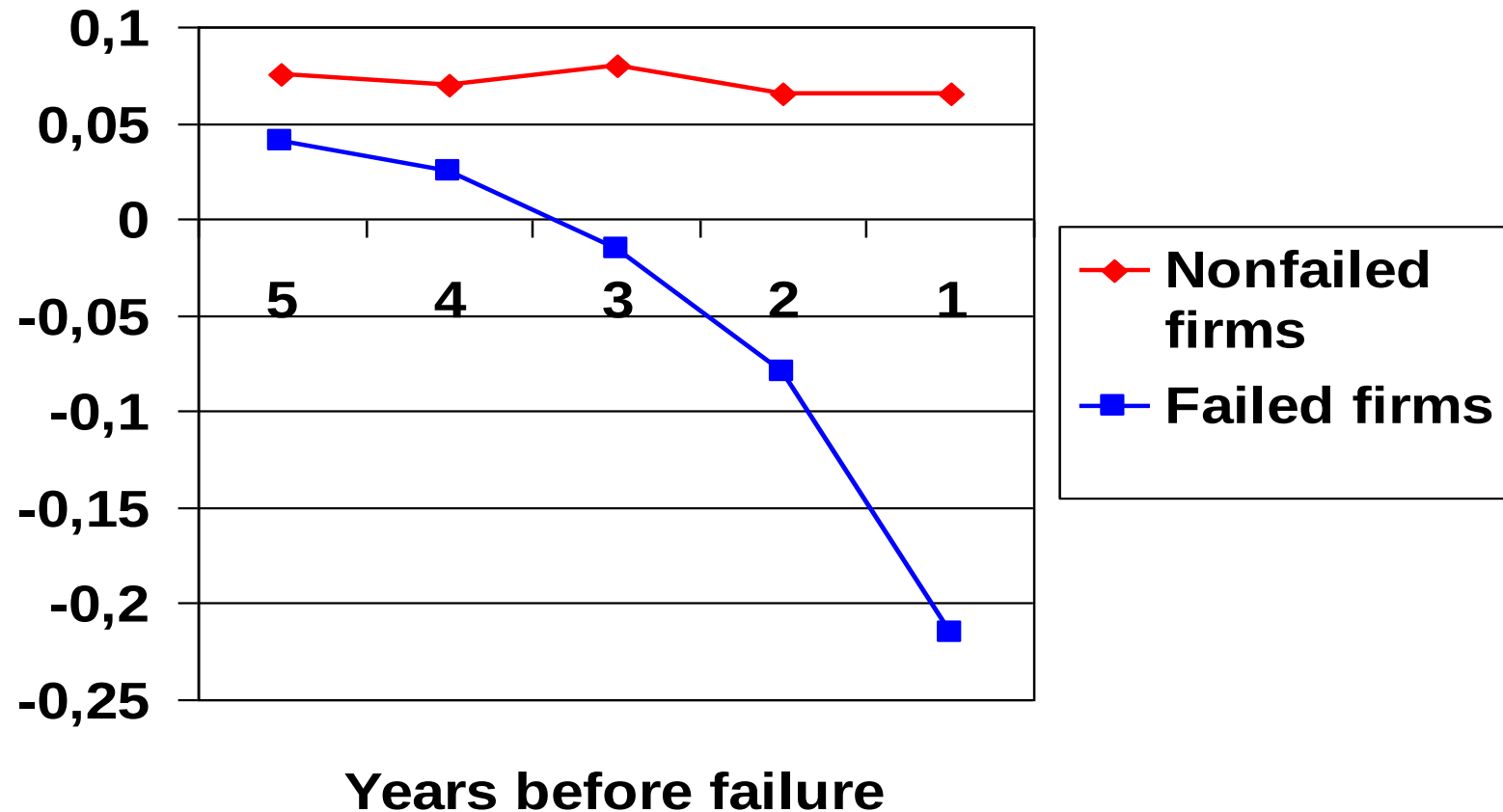
- Kone, 2013:

$$\begin{aligned}\text{INVENTORY PER SALES-\%}_{2013} &= \frac{1103,9}{6932,6} \\ &= 15,9\%\end{aligned}$$

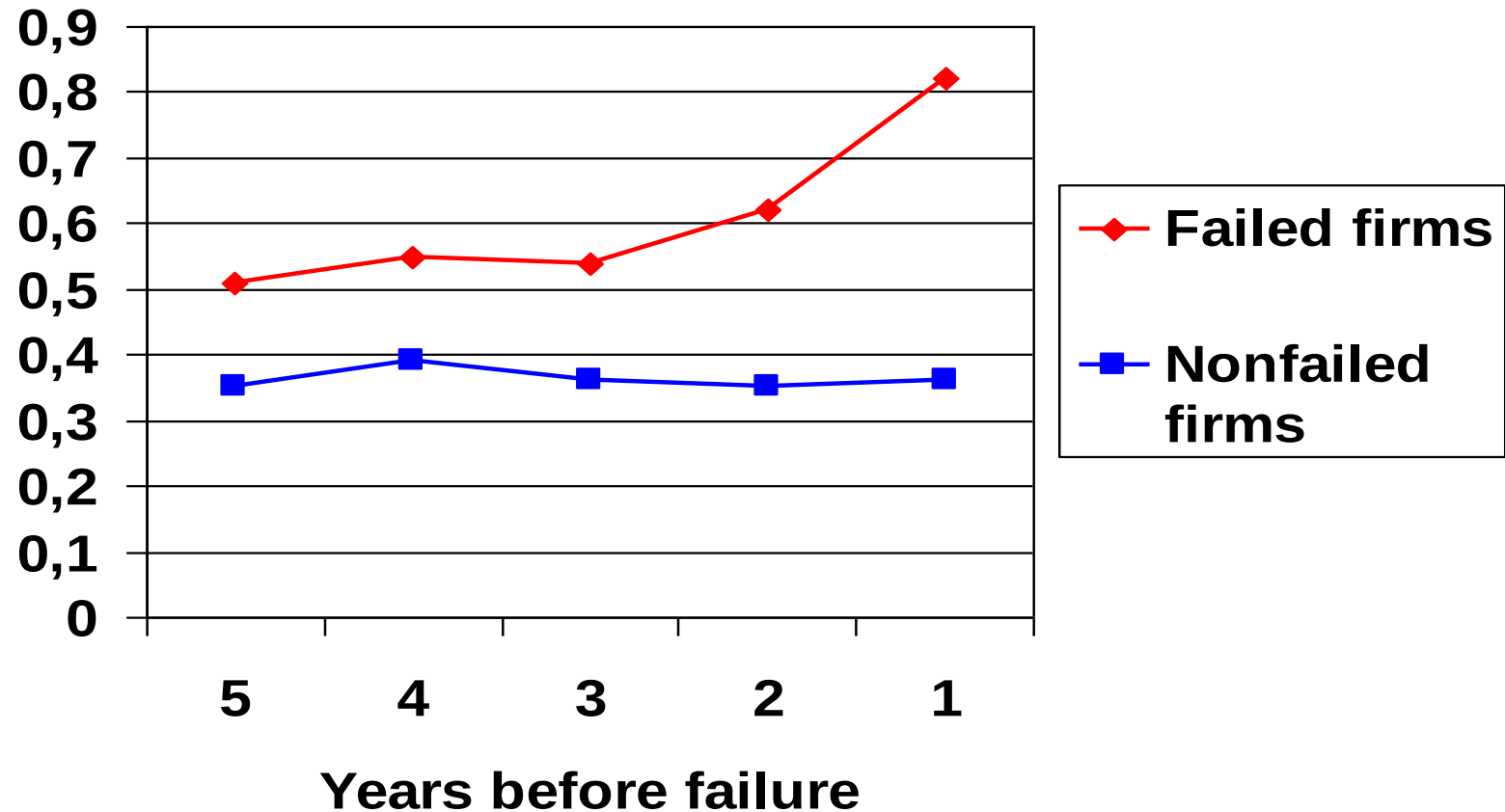
Financial ratios and credit risk

- Financial ratios **predict**
 - Profitability: Weakened earnings may launch the process that leads firms into financial distress (**early warning signs**)
 - Financial leverage: All too much debt is a signal of already-existing financial troubles (**mid-term warning signs**)
 - Liquidity and working capital: Liquidity straits are a strong signal of anticipated distress (**final warning signs**)
- Main questions are
 - How to use financial ratios in distress prediction?
 - Can we improve their prediction power by using other information?

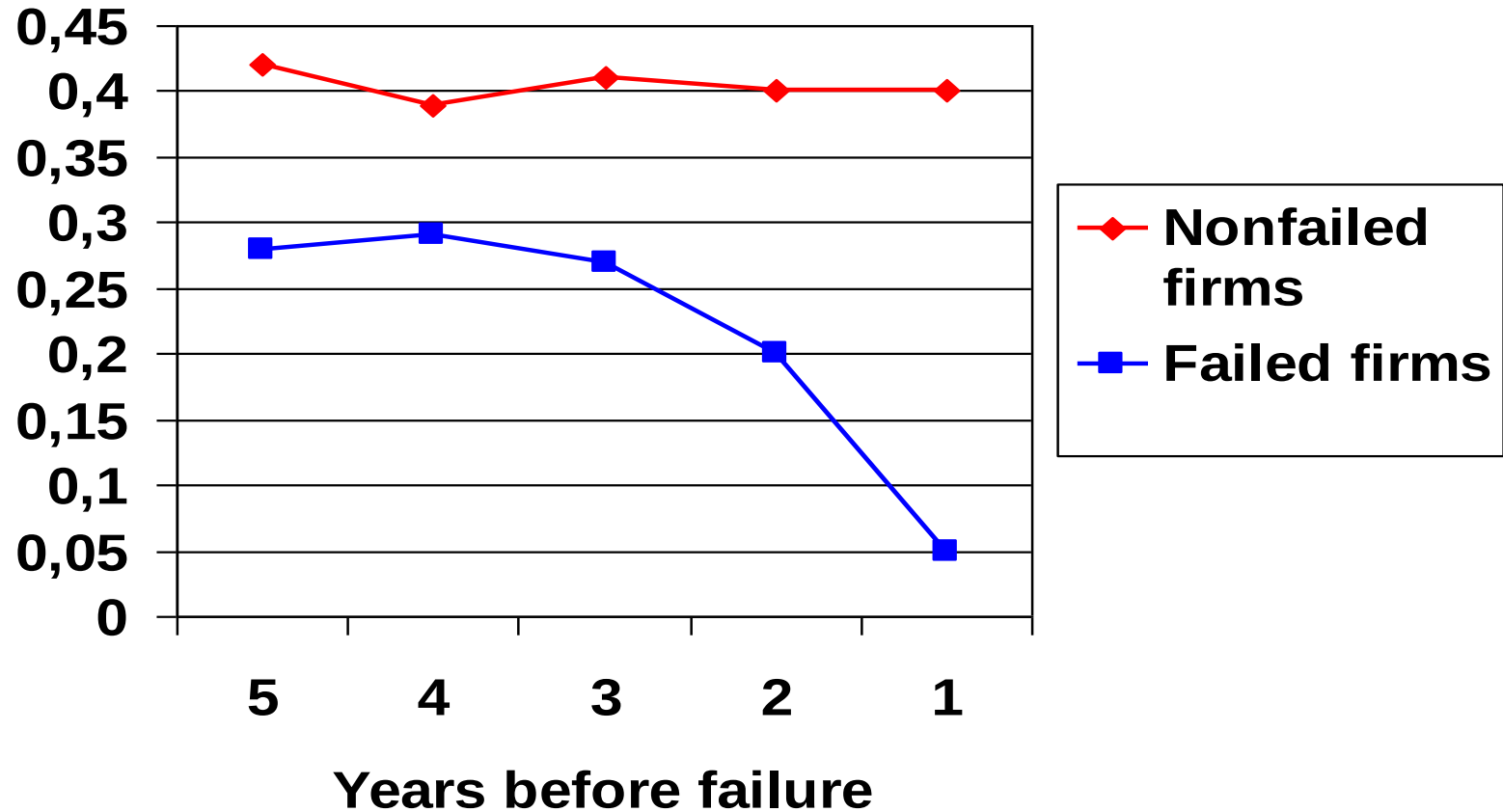
Univariate distress prediction by ROA



Univariate distress prediction by Debt to Total Assets



Univariate distress prediction by Working Capital to Total Assets



Multivariate distress prediction models

- Combine **several financial ratios**
- Multivariate models perform better than univariate models
- Classical multivariate models
 - **Altman's** model
 - **Ohlson's** model
- More recent models include other variables but financial ratios
 - Stock-market based variables
 - Executives' personal traits

Altman's Z-score for listed firms

- Original Altman's Z-score for listed firms is as follows

$$Z = 1.2X_1 + 1.4X_2 + 3.3X_3 + 0.6X_4 + 1.0X_5, \text{ where}$$

- X_1 = Working Capital / Total Assets
 - X_2 = Retained Earnings / Total Assets
 - X_3 = Earnings Before Interest and Taxes / Total Assets.
 - X_4 = Market Value of Equity / Book Value of Total Liabilities.
 - X_5 = Sales/ Total Assets.
- Zones of discrimination between the firms are
 - $Z > 2.99$ "Safe" Zone
 - $1.81 < Z < 2.99$ "Grey" Zone
 - $Z < 1.81$ "Distress" Zone

Altman's Z-score for private firms

- Original Altman's Z-score for private firms is as follows
$$Z = 0.717X_1 + 0.847X_2 + 3.11X_3 + 0.42X_4 + 0.998X_5$$
, where
 - X_1 = Working Capital / Total Assets
 - X_2 = Retained Earnings / Total Assets
 - X_3 = Earnings Before Interest and Taxes / Total Assets
 - X_4 = Book Value of Equity / Total Liabilities
 - X_5 = Sales/ Total Assets
- Zones of discrimination between the firms are
 - $Z > 2.9$ "Safe" Zone
 - $1.23 < Z < 2.9$ "Grey" Zone
 - $Z < 1.23$ "Distress" Zone

Altman's Z-score for other firms and countries

- Altman (1968) estimated the original Z-score for the sample of 66 manufacturing firms
- Since then, the model has been estimated for other industries and for firms in different countries
- Parameters of the model vary in these updates

Credit ratings

- A debt rating is an indicator of the likelihood of timely repayment of principal and interest by a borrower
- The more likely is the timely repayment, the higher is the rating
- Rating is usually expressed in terms of certain categories
- Commercial debt ratings
 - Moody's and Standard&Poors in US
 - Suomen Asiakastieto in Finland

Research evidence on distress prediction

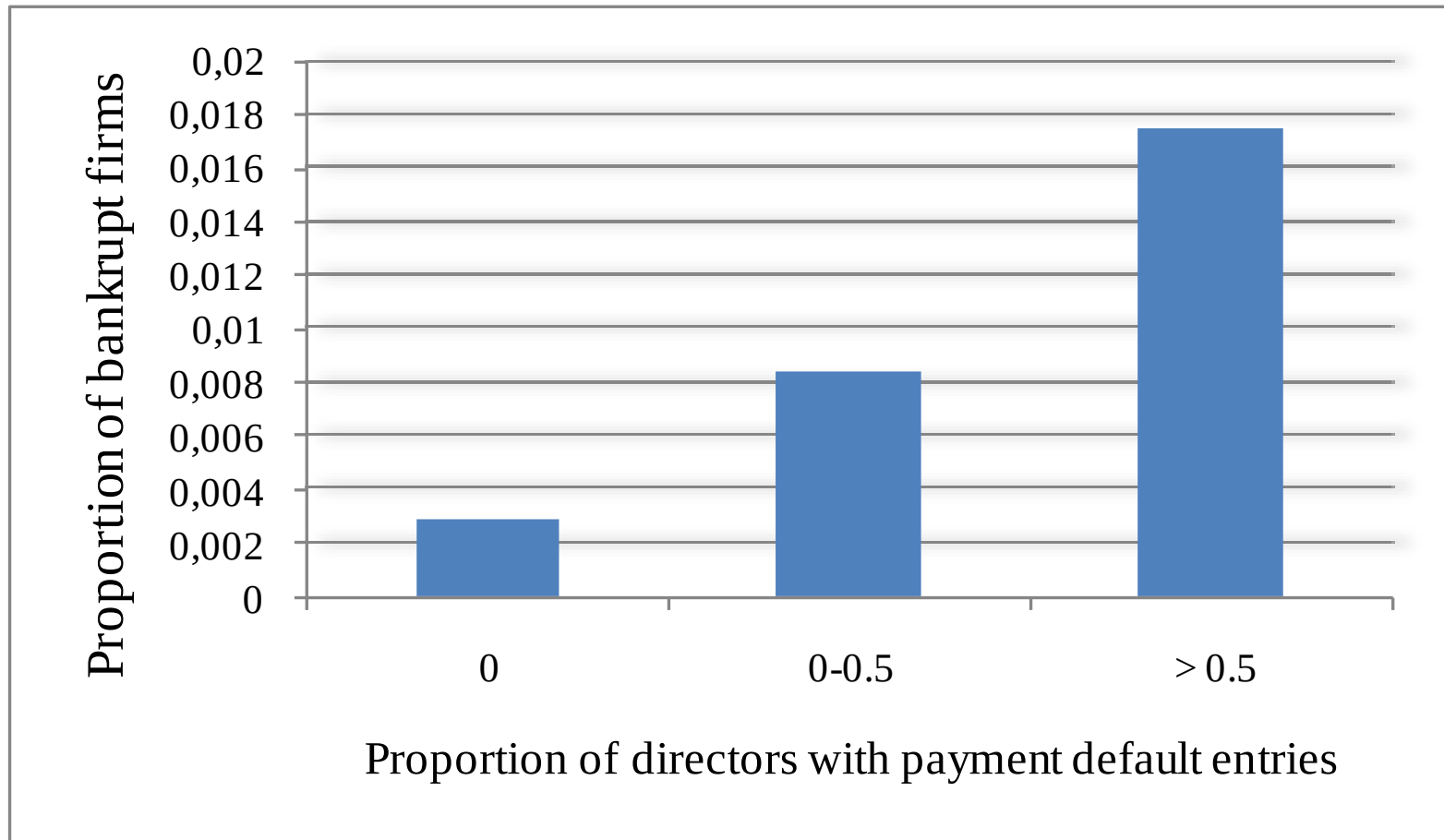
- Kallunki and Pyykkö (2013, RAST) explore whether appointing CEOs and directors with past **personal payment default entries** increases the likelihood of financial distress of the firm
 - “If managers cannot manage their own money, how could they manage the money of their company...?”
- So far, distress prediction and credit rating models have based solely on **financial ratios** or similar firm-level data
- This is the first paper to use information on managers’ **personal characteristics** in distress prediction
- Results show that it pays to check CEOs’ and directors’ personal credit defaults

Research evidence on distress prediction

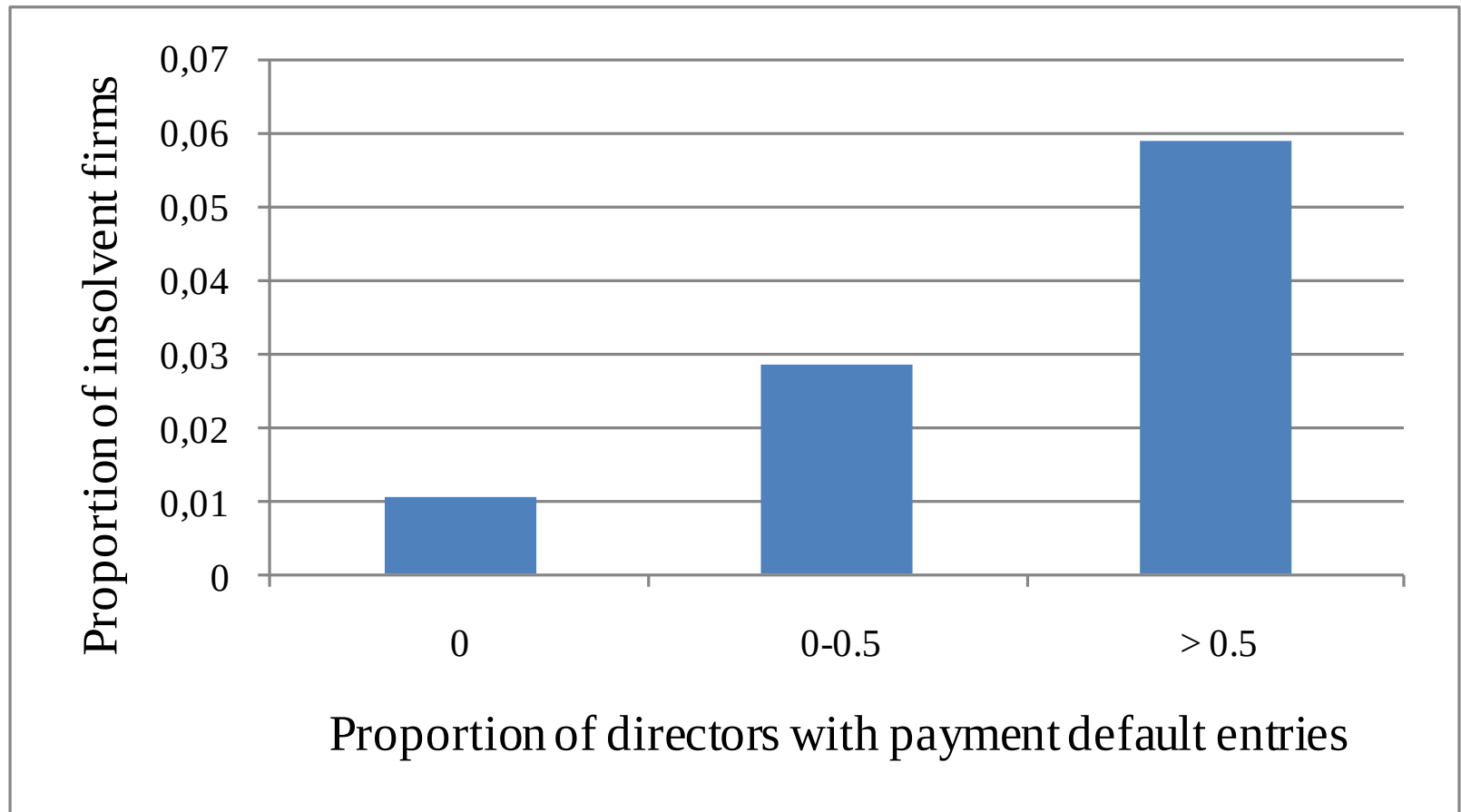
Proportion of defaulting CEOs and directors between distress and non-distress firms

	<u>Bankrupt firms</u>	<u>Insolvent firms</u>	<u>Non-distressed firms</u>
Defaulting CEOs	6.8%	5.1%	1.2%
Defaulting directors	11.9%	11.6%	2.8%

Research evidence on distress prediction



Research evidence on distress prediction



Research evidence on distress prediction

Table 8 Comparison of bankruptcy and insolvency measures in classifying firms

Category	Bankruptcy				Insolvency			
	Z-Score	Adj. Z-Score	O-Score	Adj. O-Score	Z-Score	Adj. Z-Score	O-Score	Adj. O-Score
<i>Panel B: In-sample prediction test: # of actual bankruptcy and insolvency firms</i>								
1 (Low risk)	2	1	0	0	7	6	4	4
2	0	1	2	1	20	17	17	13
3	4	4	8	6	20	14	19	18
4	7	4	4	6	22	21	27	20
5	7	7	5	7	29	26	24	23
6	7	8	12	9	32	35	23	30
7	21	22	11	12	62	55	51	48
8	24	19	12	13	85	85	77	74
9	37	37	42	37	147	151	130	129
10 (High risk)	52	58	65	70	162	176	214	227
Total	161	161	161	161	586	586	586	586

Summary

- Credit risk refers to the **risk of default** by the borrower
- Financial ratios play an important role in credit risk analysis
- Financial distress may lead to default or (costly) restructuring
- **Multivariate distress** prediction models perform better than univariate models
- Credit ratings indicate the risk of default
- Managers' personal characteristics affect default risk